Jacob Dineen

AI, Reasoning & Cognition (ARC) Lab — Arizona State University
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RESEARCH INTERESTS

I work on reasoning and alignment in large language models, with related interests in multi-agent reinforcement learning, controllability, and explainable AI (XAI). I have also held data science, ML engineering, and applied research roles across financial services and AI startups.

EDUCATION

Arizona State University — Ph.D. Artificial Intelligence *GPA 4.00/4.00 · Advisor: Ben Zhou* 2022–2027

Committee: Muhao Chen, Chitta Baral, Vivek Gupta

University of Virginia — M.Sc. Computer Science *GPA 3.96/4.00 · Advisor: Madhav Marathe* 2019–2021

Syracuse University — M.S. Data Science *GPA 4.00/4.00* 2017–2018

Grand Canyon University — B.S. Finance & Economics *GPA 3.65/4.00* 2012–2015

RESEARCH EXPERIENCE

Graduate Research Assistant, ARC Lab (Arizona State University) 2024–Present

- Research on reasoning, alignment, controllability, and multi-agent RL in LLMs.

Graduate Research Assistant, SEFCOM (Arizona State University) 2022–2024

- AI and cybersecurity research.

Applied Research, Capital One — Center for Machine Learning (C4ML) 2020–2021

- Built reinforcement learning and agent-based simulations for organizational dynamics.

Research Assistant, Biocomplexity Institute (University of Virginia) 2019–2020

- Worked on graph dynamical systems, cooperative game theory, and behavioral modeling.

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Alumbra AI Part-Time 2026

Research, Pareto AI Part-Time 2025–Present

Machine Learning Engineer, Spring Oaks Capital Full-Time 2022–2025

Data Scientist, Capital One Full-Time 2020–2022

Analyst & Business Intelligence, Real World Marketing Full-Time 2016–2019

Data Scientist, Buffalo Check LLC Part-Time 2015–2019

Optimization Analyst, Voltari Part-Time 2012–2015

PUBLICATIONS

Google Scholar Profile

+ Corresponding Author / Mentor

Peer-Reviewed Conference Proceedings (C)

C1. **Dineen, Jacob**+, RRV, Aswin, Liu, Qin, Xu, Zhikun, Ye, Xiao, Shen, Ming, Li, Zhaonan, Lu, Shijie, Baral, Chitta, Chen, Muhao, & Zhou, Ben (2025). *QA-LIGN: Aligning LLMs through Constitutionally Decomposed QA*. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 20619–20642.

C2. RRV, Aswin, **Dineen, Jacob**, Handa, Divij, Uddin, Md Nayem, Parmar, Mihir, Baral, Chitta, & Zhou, Ben (2025). *ThinkTuning: Instilling Cognitive Reflections without Distillation*. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 31236–31250.

C3. Xu, Zhikun, Shen, Ming, **Dineen, Jacob**, Li, Zhaonan, Ye, Xiao, Lu, Shijie, RRV, Aswin, Baral, Chitta, & Zhou, Ben (2025). *ToW: Thoughts of Words Improve Reasoning in Large Language Models*. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 3057–3075.

C4. **Dineen, Jacob**+, Kridel, Donald J., Dolk, Daniel R., & Castillo, David G. (2023). *Unified Explanations in Machine Learning Models: A Perturbation Approach*. In *Proceedings of the 56th Hawaii International Conference on System Sciences (HICSS-56)*, pp. 795–804.

C5. **Dineen, Jacob**+, Haque, A. S. M. Ahsan-Ul, & Bielskas, Matthew (2021a). *Formal Methods for an Iterated Volunteer's Dilemma*. In *Proceedings of the 14th International Conference on Social, Cultural, and Behavioral Modeling (SBP-BRiMS 2021)*, pp. 81–90.

C6. **Dineen, Jacob**+, Haque, A. S. M. Ahsan-Ul, & Bielskas, Matthew (2021b). *Reinforcement Learning for Data Poisoning on Graph Neural Networks*. In *Proceedings of the 14th International Conference on Social, Cultural, and Behavioral Modeling (SBP-BRiMS 2021)*, pp. 141–150.

C7. Dolk, Daniel R., Kridel, Donald J., **Dineen, Jacob**+, & Castillo, David G. (2020). *Model Interpretation and Explainability towards Creating Transparency in Prediction Models*. In *Proceedings of the 53rd Hawaii International Conference on System Sciences (HICSS-53)*.

Peer-Reviewed Workshop Papers (WS)

WS1. Srinivasan, Adarsh, **Dineen, Jacob**+, Sarfraz, Muhammad Uzair, Afzal, Muhammad Umar, Riaz, Irbaz, Zhou, Ben (2026). *RECAP: Transparent Inference-Time Emotion Alignment for Medical Dialogue Systems*. In *Proceedings of the 8th Workshop on Clinical Natural Language Processing (Clinical NLP) @ LREC 2026*, pp. 350–368. ORAL

WS2. Li, Zhaonan, Chickering, Kyle R., Li, Bangzheng, **Dineen, Jacob**, Ye, Xiao, Xu, Zhikun, Lu, Shijie, Huang, Yuxi, Shen, Ming, Nguyen, Bach, Pavuluri, Jaya Adithya, Nguyen, Mau Son, Chavan, Sanika, Le, Ngoc Minh Thu, Chen, Muhao, Zhou, Ben (2026). *VisAnalog: A Diagnostic Suite for Visual Concept Transfer on Natural Images*. CVPR Workshop on Visual Concepts (VisCon), 2026. ORAL

Working Papers / Under Review (W)

W1. Li, Zhaonan, Lu, Shijie, Wang, Fei, **Dineen, Jacob**, Ye, Xiao, Xu, Zhikun, Liu, Siyi, Cho, Young Min, Li, Bangzheng, Chang, Daniel, Nguyen, Kenny, Yang, Qizheng, Chen, Muhao, Zhou, Ben (2025). *Unbiased Visual Reasoning with Controlled Visual Inputs*. Under Review, EMNLP 2026

W2. Liu, Qin, **Dineen, Jacob**, Huang, Yuxi, Zhang, Sheng, Poon, Hoifung, Xiao, Chaowei, Zhou, Ben, Chen, Muhao (2025). *ArenaBencher: Automatic Benchmark Evolution via Multi-Model Competitive Evaluation*. Under Review, EMNLP 2026

W3. Ye, Xiao, Li, Zhaonan, **Dineen, Jacob**, Xu, Zhikun, Lu, Shijie, Shen, Ming, Shrivastava, Shaswat, Ahuja, Avneet, Zhou, Ben (2025). *Learning Verifiable Reasoning Programs through Cohort Consistency*. Under Review, EMNLP 2026

W4. Shen, Ming, Xu, Zhikun, Ye, Xiao, **Dineen, Jacob**, Zhou, Ben (2025). *BOW: Training Language Models to Reason Over Plausible Next Words*. Under Review, EMNLP 2026

W5. RRV, Aswin, **Dineen, Jacob**, Handa, Divij, Parmar, Mihir, Zhou, Ben, Baral, Chitta, Mishra, Swaroop (2026). *Mid-training with Self-Generated Data Improves Reinforcement Learning in Language Models*. Under Review, NeurIPS 2026

W6. **Dineen, Jacob**, RRV, Aswin, Xu, Zhikun, Zhou, Ben (2026). *Vocabulary Dropout for Curriculum Diversity in LLM Co-Evolution*. Under Review, COLM 2026

W7. Xu, Zhikun, Feng, Yu, **Dineen, Jacob**, Shi, Taiwei, Zhao, Jieyu, Zhou, Ben (2026). *Skill Reuse as Compression in Agentic RL*. Under Review, EMNLP 2026

W8. Ye, Xiao, **Dineen, Jacob**, Zhu, Evan, Lu, Shijie, Song, Kevin, Zhou, Ben (2026). *HINDCAST: Replaying Prediction Markets to Evaluate LLM Forecasters*. Under Review, EMNLP 2026

W9. Ye, Xiao, **Dineen, Jacob**, Li, Zhaonan, Xu, Zhikun, Chen, Weiyu, Lu, Shijie, Huang, Yuxi, Shen, Ming, Tran, Phu, Yum, Ji-Eun Irene, Khan, Muhammad Ali, Afzal, Muhammad Umar, Riaz, Irbaz Bin, Zhou, Ben (2025). *Evaluating Medical LLMs by Levels of Autonomy: A Survey*. In Preparation

SKILLS

LLM Methods	pre/mid/post-training, RL (GRPO, RLHF), reward modeling, self-play, agentic tool use, RAG, synthetic data generation, evaluation and benchmarking
ML Libraries	PyTorch, TensorFlow, JAX, Polars, Dask, HuggingFace, TRL, vLLM, VeRL, PySpark, NetworkX, DGL, Torch Geometric, BoTorch, SnowparkML, Anthropic/OpenAI SDKs
Languages	Python, Rust, C, C++, Java, JavaScript, R, Bash, PRISM, x86-64
Databases	MySQL, PostgreSQL, MongoDB, Snowflake, Redis, SQLite, Redshift
Tools	Git, GitHub Actions, Docker, Kubernetes, Helm, Airflow, AWS (ECS, ECR, S3, EKS, CodeBuild), SageMaker, Databricks, FastAPI, Next.js, pytest, OR-Tools, Sigma, Streamlit

SERVICE

Conference Reviewer: HICSS, SBP-BRiMS, NAACL, EMNLP, EACL, COLM

Teaching / Tutoring:

- CGCC Calculus & Linear Algebra Tutor (2019)
- Teaching Assistant — ASU CSE 365 / pwn.college (Security)

MISCELLANEOUS

Expert AI Trainer, Pareto AI (2024–Present)

Independent Contractor, Mercor Intelligence (2025–Present)

Cybersecurity: pwn.college green belt (binary exploitation), user: `jdin`

GRADUATE COURSEWORK

Arizona State University — PhD Computer Science:
Knowledge Representation; Computer Systems Security; Software Security; Planning and Learning Methods in AI; Algorithms; Research/Dissertation Hours.

University of Virginia — MSc Computer Science:
Algorithms; Machine Learning; Computer Vision; Formal Methods; Reinforcement Learning; Graph Mining; Learning Theory (Game Theory); Cloud Computing; Research Hours.

Syracuse University — MS Data Science:
Data Analysis and Decision Making; Business Analytics; Financial Analytics; Marketing Analytics; Advanced Information Systems; Data Science; Data Warehousing; Text Mining; Scripting for Data Analysis; Information Policy.